6.3.4 System integration and functional operation

Integration turns components into dependable, certifiable systems through engineering disciplines, verification, and operational doctrine:

- *Design-for-assurance*: Co-design of hardware and software with formal methods where feasible; component interfaces standardized to enable modular certification and incremental upgrades without system regression.
- *Pre-mission workflow*: Mission intent and constraints are uploaded and verified against policy guardrails; planners generate candidate strategies, which are vetted in a digital twin (closed-loop simulation) before authorization.
- *Launch and comms layering*: Multi-layer connectivity established (LOS, mesh, SATCOM, laser link). The autonomy stack selects the optimal link based on latency, bandwidth, and threat model; QoS policies prioritize command, control, and critical telemetry.
- *On-station behavior and cooperative tactics*: Vehicles execute locally optimized plans while sharing summarized state and intent. Swarm coordination uses leaderless consensus or hybrid leader-follower modes depending on mission type; distributed task allocation optimizes utility (coverage, time-to-target, risk).
- *Decision escalation and human governance*: Actions exceeding predefined thresholds (lethality, high collateral risk, sensitive geography) generate explainable alerts and require human authorization (human-inthe-loop). Lower-risk adaptive actions may proceed autonomously with post-fact review.
- *Resilience operations*: Upon detection of jamming, spoofing, or subsystem faults, the FDIR module invokes contingency plans, e.g.,